

Program 3

```
# include < iostream >
# include < string >
using namespace std ;

class Product {
private :
int productId ;
string productName ;
float price ;
int monthlySales [12];
int totalQuantity ;
float totalBill ;

public :
void acceptDetails () {
cout << " Enter Product ID: ";
cin >> productId ;
cin . ignore () ;
cout << " Enter Product Name : ";
getline ( cin , productName ) ;
cout << " Enter Price per unit : ";
cin >> price ;
cout << " Enter monthly sales for 12 months : ";
totalQuantity = 0;
for (int i = 0; i < 12; i ++ ) {
cin >> monthlySales [ i ];
totalQuantity += monthlySales [ i ];
}
totalBill = totalQuantity * price ;
}

void displayDetails () {
cout << "\n Product ID: " << productId << endl ;
cout << " Product Name : " << productName << endl ;
cout << " Price : " << price << endl ;
cout << " Total Quantity Sold : " << totalQuantity << endl ;
cout << " Total Bill : " << totalBill << endl ;
}

float getTotalBill () const {
return totalBill ;
}
};

int main () {
int n ;
cout << " Enter number of products : ";
```

```

cin >> n ;
49
50
Product products [100];
51
float grandTotal = 0;
52
53
for (int i = 0; i < n ; i ++ ) {
54
cout << "\n- - - Product " << ( i +1) << " Details ---" <<
endl ;
55
products [ i ]. acceptDetails () ;
56
grandTotal += products [ i ]. getTotalBill () ;
57
}
58
59
cout << "\n- - - - All Products Details -----" << endl ;
60
for (int i = 0; i < n ; i ++ ) {
61
products [ i ]. displayDetails () ;
62
}
63
64
cout << "\ nGrand Total Bill : " << grandTotal << endl ;
65
return 0;
}

```